

A Lightweight Adaptive Multimodal 3D CNN for ADHD Classification Using Structural MRI and Phenotypic Features

Khushi Jindal

Department of Information Technology, Delhi Technological University, Delhi, India
khushijindal06@gmail.com

Abstract. Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder whose diagnosis primarily relies on behavioral and clinical assessment. Machine learning methods based on neuroimaging have shown potential for supporting automated ADHD classification; however, many deep learning approaches employ increasingly complex architectures with substantial computational and parameter requirements. This work presents *ADHDNet*, a lightweight multimodal 3D convolutional neural network for ADHD classification using structural magnetic resonance imaging (sMRI) and phenotypic information. The proposed framework employs depthwise-separable 3D convolutional blocks to reduce model parameters while retaining volumetric feature extraction capability. A three-dimensional efficient channel attention mechanism recalibrates informative feature channels, while an adaptive gated fusion module dynamically combines MRI-derived and phenotypic representations. Experiments were conducted on a multisite subset of the ADHD-200 dataset comprising 497 subjects after excluding a site without ADHD cases. Structural MRI volumes were resized to $64 \times 64 \times 64$, normalized, and combined with five phenotypic features. The proposed model contains only 71,558 trainable parameters (0.0716M). On a stratified held-out validation set, ADHDNet achieved 89.00% accuracy and an AUC of 0.9535, with a sensitivity of 93.33%, specificity of 85.45%, precision of 84.00%, and F1-score of 0.88. Compared with a standard Conv3D multimodal comparator containing 1.376M parameters, ADHDNet reduces the parameter count by approximately 94.8%. The results demonstrate that a compact multimodal 3D architecture can provide effective ADHD classification while substantially reducing model complexity.

Keywords: ADHD · Structural MRI · Multimodal learning · 3D CNN · Depthwise-separable convolution · Efficient channel attention · Lightweight deep learning

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by persistent patterns of inattention, hyperactivity, and impulsivity. Diagnosis is primarily based on behavioral assessment and clinical

evaluation, which can be challenging because symptoms may overlap with those of other developmental and psychiatric conditions. Consequently, neuroimaging-based machine learning has been investigated as a complementary computational approach for identifying patterns associated with ADHD [4,8,9].

Magnetic resonance imaging (MRI), particularly structural MRI (sMRI), provides non-invasive information about brain anatomy and has been extensively investigated for automated ADHD analysis [10]. Conventional machine learning approaches have extracted regional gray-matter volume, cortical thickness, and personal characteristics before applying classifiers such as support vector machines and random forests. For example, Lohani and Rana [1] used structural MRI-derived features together with personal characteristics and reported a maximum classification accuracy of 75% using conventional machine learning models on an age-balanced ADHD dataset.

Deep learning approaches have subsequently enabled direct learning of imaging representations. Recent studies have explored convolutional neural networks, multimodal architectures, attention mechanisms, and ensemble approaches for ADHD analysis [5,6,7,3]. These approaches demonstrate the potential of learned neuroimaging representations but can introduce substantial computational requirements.

At the same time, model efficiency has become increasingly relevant for neuroimaging applications. Deepika et al. [2] specifically investigated parameter-efficient ADHD classification and highlighted the computational cost associated with increasingly complex deep learning architectures. Their lightweight attention-driven KAN framework reported 79.25% accuracy while using only a few thousand parameters. This motivates the development of compact deep learning architectures that can process volumetric MRI while incorporating complementary phenotypic information without unnecessarily increasing model complexity.

In this work, we propose *ADHDNet*, a lightweight adaptive multimodal 3D CNN for ADHD classification. Instead of increasing network depth and width, the proposed architecture reduces parameter requirements through depthwise-separable 3D convolutions. A 3D efficient channel attention mechanism is incorporated to improve channel-wise feature recalibration, and an adaptive gated fusion module dynamically combines MRI and phenotypic representations. The main contributions of this work are:

1. A **lightweight 3D MRI encoder** based on depthwise-separable Conv3D operations for parameter-efficient volumetric feature extraction.
2. Integration of a **3D efficient channel attention (ECA)** mechanism for channel-wise feature recalibration with limited additional parameters.
3. An **adaptive gated multimodal fusion mechanism** for combining structural MRI and phenotypic representations.
4. Evaluation on a **multisite subset of the ADHD-200 dataset** using structural MRI and phenotypic information.
5. Explicit analysis of **parameter efficiency**, including comparison with a conventional Conv3D multimodal architecture.

2 Related Work

MRI-Based ADHD Classification. MRI-based machine learning has been widely investigated for ADHD classification [4,8,9]. Earlier approaches generally relied on manually extracted neuroanatomical or functional connectivity features followed by conventional classifiers. Lohani and Rana [1] investigated structural MRI together with personal characteristic information and extracted gray-matter volume and cortical thickness features; their best configuration achieved 75% accuracy using conventional classifiers and repeated ten-fold crossvalidation. More recent studies have increasingly adopted deep learning to learn representations directly from neuroimaging data. CNN-based approaches have been investigated for fMRI-based ADHD classification [5], while ensemble CNN approaches have been explored for distinguishing ADHD subtypes. Pedrollo et al. [3] reported CNN ensemble-based classification using resting-state fMRI in a cohort of 623 participants. These studies demonstrate the potential of deep neural networks for learning discriminative neuroimaging representations; however, many focus primarily on improving predictive performance through increasingly complex architectures.

Multimodal ADHD Analysis. Phenotypic variables can provide complementary information to neuroimaging measurements. Age, sex, medication status, and clinical characteristics may contain information that is not directly represented in image features. The study by Lohani and Rana [1] provides evidence for the usefulness of combining structural MRI-derived characteristics with personal information. This motivates multimodal architectures that learn an imaging representation while independently encoding phenotypic information before combining both modalities.

Lightweight Deep Learning. Model complexity is an important consideration when deep learning systems are intended for computationally constrained environments. Deepika et al. [2] explicitly investigated parameter efficiency for ADHD classification and proposed an attention-driven KAN architecture with only a few thousand parameters. Therefore, rather than treating model size as secondary, the present study considers *parameter efficiency as a primary design objective*. ADHDNet uses depthwise-separable 3D convolution to reduce the number of learnable parameters while retaining volumetric processing capability.

3 Materials and Methods

3.1 Dataset

Experiments were conducted using the preprocessed anatomical component of the publicly available ADHD-200 dataset. The dataset initially contained 575 successfully processed subjects (353 controls and 222 ADHD) distributed across six sites: KKI, NYU, OHSU, Peking_1, Pittsburgh, and NeuroIMAGE. The

Table 1. Dataset distribution after site filtering.

Site	ADHD	Control	Total
KKI	22	60	82
NYU	115	96	211
OHSU	36	35	71
Peking_1	24	61	85
NeuroIMAGE	25	23	48
Total	222	275	497

Pittsburgh site contained 78 control subjects but no ADHD subjects. Because a site containing only one class cannot contribute meaningful within-site binary discrimination, it was excluded from model development. The final cohort therefore consisted of 497 subjects (222 ADHD and 275 controls). The distribution after site exclusion is shown in Table 1.

3.2 MRI Preprocessing

The structural MRI data were provided in NIfTI format. To obtain a computationally manageable input representation, each MRI volume was resized to $64 \times 64 \times 64$ voxels using trilinear interpolation. Each resulting volume was independently normalized using min-max normalization,

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (1)$$

such that voxel intensities were mapped to the range $[0, 1]$. The present implementation intentionally uses a relatively simple preprocessing pipeline: no additional skull stripping, atlas-based segmentation, N4 bias correction, or anatomical segmentation was introduced, in order to maintain a compact and reproducible experimental pipeline.

3.3 Phenotypic Features

Five phenotypic variables were incorporated into the multimodal framework: age, sex, medication status, site, and ADHD Index. Missing values represented by -999 were converted to missing values before imputation. Continuous variables were normalized, while categorical variables were numerically encoded. The phenotypic branch therefore produces a compact representation complementary to the MRI-derived representation.

4 Proposed ADHDNet

ADHDNet consists of three major components: (i) a lightweight 3D MRI encoder, (ii) a phenotypic feature encoder, and (iii) adaptive gated multimodal fusion followed by classification. The overall architecture is illustrated in Fig. 1.

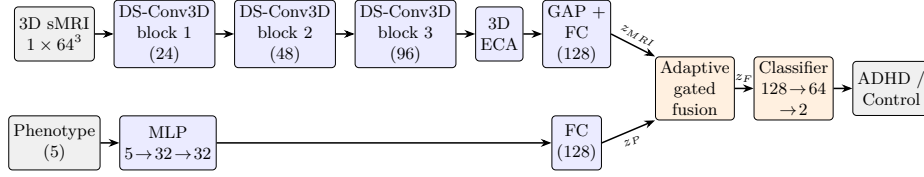


Fig. 1. Overview of ADHDNet. The sMRI branch uses three depthwise-separable (DS) Conv3D blocks followed by 3D efficient channel attention (ECA), global average pooling (GAP), and projection to a 128-dimensional embedding. The phenotypic branch uses a compact MLP. Both embeddings are combined by adaptive gated fusion before classification.

4.1 Lightweight 3D MRI Encoder

The imaging branch uses three depthwise-separable 3D convolutional blocks with channel progression $1 \rightarrow 24 \rightarrow 48 \rightarrow 96$ and spatial downsampling after each block. Depthwise-separable convolution decomposes a conventional convolution into (i) a depthwise spatial convolution and (ii) a pointwise $1 \times 1 \times 1$ convolution. For a conventional 3D convolution with kernel size K , the parameter count is approximately

$$P_{\text{conv}} = K^3 C_{in} C_{out}, \quad (2)$$

whereas depthwise-separable convolution requires approximately

$$P_{\text{DS}} = K^3 C_{in} + C_{in} C_{out}. \quad (3)$$

Since the second expression can be considerably smaller when C_{out} is large, depthwise-separable convolution provides a mechanism for reducing the number of trainable parameters. After the three convolutional blocks, adaptive global average pooling produces a 96-dimensional representation, and a fully connected projection maps it to a 128-dimensional MRI embedding.

4.2 Efficient Channel Attention

To improve feature recalibration without introducing a large parameter overhead, a three-dimensional efficient channel attention mechanism is incorporated. The attention operation estimates channel importance from globally pooled feature responses and applies channel-wise weights to the volumetric feature maps:

$$F_{att} = A(F) \odot F, \quad (4)$$

where F represents the feature tensor and $A(F)$ the learned channel attention weights. This allows the network to emphasize informative feature channels while maintaining the lightweight nature of the architecture.

4.3 Phenotypic Encoder

The phenotypic input consists of five features. The phenotype encoder is a small multilayer perceptron with dimensions $5 \rightarrow 32 \rightarrow 32$. ReLU activation, batch normalization, and dropout are used to regularize the representation. The resulting 32-dimensional phenotype representation is projected to 128 dimensions before multimodal fusion.

4.4 Adaptive Gated Fusion

Simple concatenation treats both modalities equally at the fusion stage. Instead, ADHDNet employs a learnable gating mechanism. Let $z_{MRI} \in \mathbb{R}^{128}$ denote the MRI representation and $z_P \in \mathbb{R}^{128}$ the projected phenotype representation. The two representations are first concatenated, $z = [z_{MRI}; z_P]$, and a gating network generates

$$g = \sigma(W_2 \delta(W_1 z)), \quad (5)$$

where σ denotes the sigmoid function and δ a nonlinear activation. The fused representation is then

$$z_F = g \odot z_{MRI} + (1 - g) \odot z_P. \quad (6)$$

Thus, the model can dynamically determine the relative contribution of the imaging and phenotypic representations for each sample.

4.5 Classification Head

The fused 128-dimensional representation is passed through a compact classifier with dimensions $128 \rightarrow 64 \rightarrow 2$, whose two output units correspond to the ADHD and control classes. Softmax probabilities are used to obtain the predicted class.

5 Model Complexity

Parameter efficiency is a central design consideration of ADHDNet. The proposed model contains 71,558 trainable parameters (≈ 0.0716 M). For comparison, a standard Conv3D multimodal architecture implemented under the same experimental framework contains 1,376,080 parameters (≈ 1.376 M). The relative reduction is

$$\frac{1,376,080 - 71,558}{1,376,080} \times 100 \approx 94.8\%. \quad (7)$$

Table 2 summarizes the comparison. The parameter count demonstrates that the proposed architecture substantially reduces the number of trainable weights without removing the multimodal component.

Table 2. Model complexity comparison.

Model	MRI encoder	Parameters	Relative
Standard Conv3D multimodal	Conventional Conv3D	1.376M	$19.23 \times$
ADHDNet	DS-Conv3D + ECA	0.0716M	$1 \times$

6 Experimental Setup

Training Strategy. The network was trained using the AdamW optimizer with a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , batch size of 8, a maximum of 150 epochs, early stopping with patience 15, label smoothing of 0.1, gradient clipping at 1.0, and cosine annealing learning-rate scheduling. Weighted cross-entropy loss was used to account for class imbalance. The random seed was fixed to 42; Python, NumPy, and PyTorch random states were seeded, and deterministic CUDA settings were used where applicable.

Experimental Protocol. The final dataset was divided using a stratified 80:20 train-validation split, giving 397 training subjects and 100 held-out validation subjects (45 ADHD and 55 controls). Normalization statistics were calculated from the training data and applied to the validation data. Importantly, the validation set was used for early stopping and model selection; therefore, the reported results should be interpreted as *held-out validation performance* rather than independent external test performance.

Evaluation Metrics. With ADHD as the positive class, the model was evaluated using

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Sensitivity} = \frac{TP}{TP + FN}, \quad (8)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad (9)$$

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (10)$$

Receiver operating characteristic (ROC) analysis was also performed and the area under the ROC curve (AUC) was calculated.

7 Results

7.1 Classification Performance

Table 3 reports the performance of ADHDNet on the held-out validation set, and Table 4 gives the corresponding confusion matrix. The model correctly classified 47 of 55 controls and 42 of 45 ADHD subjects; 8 controls were misclassified

Table 3. Validation performance of ADHDNet.

Metric	ADHDNet
Accuracy	89.00%
AUC	0.9535
Sensitivity	93.33%
Specificity	85.45%
Precision	84.00%
NPV	94.00%
F1-score	0.88

Table 4. Confusion matrix on the validation set ($n = 100$).

		Predicted	
		ADHD	Control
Actual	ADHD	42 (TP)	3 (FN)
	Control	8 (FP)	47 (TN)

as ADHD and 3 ADHD subjects as controls. The relatively high sensitivity indicates that most ADHD subjects in the validation set were correctly identified, while the specificity indicates that the model also correctly rejected a substantial proportion of control subjects.

7.2 ROC Analysis

The proposed model achieved an AUC of 0.9535, indicating strong discrimination on the held-out validation cohort. However, because the validation set was used for early stopping and model selection, the reported AUC should not be interpreted as an independent external validation result.

7.3 Parameter Efficiency

The most important characteristic of ADHDNet is its compact parameter count: 0.0716M parameters compared with 1.376M for the standard Conv3D comparator, i.e., approximately 94.8% fewer parameters. The proposed architecture therefore demonstrates that multimodal 3D MRI analysis does not necessarily require a large number of trainable parameters.

7.4 Comparison with Existing Studies

The objective of the literature comparison is not to rank models based solely on reported accuracy, because the studies use different datasets, modalities, preprocessing pipelines, class definitions, and validation strategies. Lohani and Rana [1] used structural MRI-derived features and personal characteristics with conventional machine learning and reported 75% maximum accuracy. Deepika et al. [2] investigated parameter-efficient ADHD classification and reported 79.25% accuracy using a lightweight attention-driven KAN framework with only a few thousand parameters. Pedrollo et al. [3] used resting-state fMRI from 623 participants and reported $97.6 \pm 2.1\%$ accuracy for ADHD diagnosis using an ensemble approach. These results demonstrate the diversity of existing approaches; direct numerical comparison is inappropriate because the underlying modalities and

Table 5. Comparison with selected recent ADHD studies. Accuracies are not directly comparable due to differing modalities, datasets, and protocols.

Study	Year	Modality	Main approach	Parameters	Accuracy
Lohani & Rana [1]	2023	sMRI + personal char.	Conventional ML	Not reported	75.0%
Deepika et al. [2]	2025	ADHD-200-derived features	Lightweight attention + KAN	Few thousand	79.25%
Pedrollo et al. [3]	2026	rs-fMRI	CNN ensemble	Not primary focus	97.6 $\pm$ 2.1%
ADHDNet (ours)	2026	3D sMRI + phenotype	DS-Conv3D + ECA + gated fusion	0.0716M	89.0%

experimental protocols differ (Table 5). ADHDNet differs from these studies by explicitly combining volumetric structural MRI, phenotypic information, and a parameter-efficient 3D convolutional architecture.

8 Discussion

The results demonstrate the feasibility of using a compact 3D CNN to integrate structural MRI and phenotypic information for ADHD classification. A key design decision was to avoid increasing model capacity through deeper conventional convolutional layers; instead, depthwise-separable convolution was used to reduce the parameter requirements of volumetric feature extraction. The proposed model contains only 71,558 parameters while maintaining an AUC of 0.9535 on the held-out validation set, a 94.8% reduction in parameters relative to the standard Conv3D comparator implemented under the same experimental framework.

The phenotypic information provides a complementary modality to the MRI representation. Rather than simply concatenating the two representations, the adaptive gating mechanism allows the model to generate a sample-dependent fusion representation. The ECA module provides channel-wise recalibration while adding relatively little architectural complexity, so the architecture remains compact while incorporating attention-based feature refinement.

The parameter-efficiency focus is particularly relevant because recent ADHD literature has begun to explicitly investigate lightweight architectures [2]. The present work positions ADHDNet within this emerging direction while focusing specifically on 3D structural MRI and phenotypic multimodal learning. It is important, however, not to interpret the 89% accuracy as evidence that ADHDNet is universally superior to previously published approaches, since existing studies use different datasets, modalities, preprocessing procedures, class definitions, and validation protocols. The main contribution of the present study is consequently the *combination of multimodal volumetric processing and parameter efficiency*, rather than achieving the highest reported ADHD classification accuracy.

Planned Ablation Analysis. To further validate the contribution of individual components, the following ablation experiments are planned: (i) an MRI-only model, removing the phenotypic branch; (ii) MRI + ECA, evaluating the contribution of efficient channel attention; (iii) MRI + phenotype with simple concatenation in place of adaptive gated fusion; (iv) the lightweight multimodal

architecture without ECA; and (v) the complete ADHDNet. No numerical ablation results are reported here until those experiments are completed.

Limitations. Several limitations should be considered. First, the present study uses a single held-out validation split rather than an independent external test cohort; the reported results should therefore not be interpreted as evidence of external generalization. Second, although the dataset contains multiple acquisition sites, the current experiment does not explicitly evaluate cross-site generalization. Third, the MRI preprocessing pipeline is intentionally simple and does not include skull stripping, registration, or atlas-based segmentation. Fourth, the current work focuses on binary ADHD-versus-control classification and does not investigate ADHD subtypes. Finally, the lightweight claim is currently supported primarily through parameter count; future evaluation should additionally report FLOPs, inference latency, peak memory consumption, and model storage requirements under a standardized hardware and software environment.

9 Conclusion

This paper presented ADHDNet, a lightweight adaptive multimodal 3D CNN for ADHD classification using structural MRI and phenotypic information. The architecture combines depthwise-separable 3D convolution, efficient channel attention, a compact phenotypic encoder, and adaptive gated multimodal fusion. Using a multisite subset of the ADHD-200 dataset containing 497 subjects, the proposed model achieved 89.00% accuracy and an AUC of 0.9535 on a held-out validation set. More importantly, ADHDNet contains only 71,558 trainable parameters, an approximately 94.8% reduction relative to the standard Conv3D comparator developed under the same experimental framework. The results indicate that a compact architecture can effectively integrate volumetric structural MRI and phenotypic information without relying on a large number of trainable parameters. Future work will focus on systematic ablation studies, standardized computational profiling, cross-site evaluation, and independent external validation.

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